

Zihan Hei

Statistics Undergraduate Student

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 [GitHub](#)
[Quarto.pub](#)
[Personal academic website](#)

Education

AUG 2023 –
JUN 2027

Bachelor of Arts in Statistics

Binghamton University, State University of New York

Minor in Digital and Data Studies | GPA: **3.8**/4.0

Research Experience

NOV 2025 –
PRESENT

Transcriptomic Biomarker Discovery and Validation in Chemo-Immunotherapy

Independent Study with Dr. Nancy Guo | Binghamton University

- Reproduce and validate RNA-seq and single-cell RNA-seq preprocessing workflows, including quality control, normalization, gene filtering, and highly variable gene selection.
- Apply NMF, differential expression analysis, cross-study comparisons, and GSEA to identify and validate treatment-response biomarkers and biological pathways (manuscript in preparation).

AUG 2025 –
PRESENT

Machine Learning for Health Information Avoidance

Independent Study with Dr. Shane McCarty and Dr. Heather Orom | Binghamton University

- Clean, recode, and score sociodemographic, psychological, health behavior, and media-use variables in R.
- Develop and compare regression, random forest, and MARS models using cross-validation and predictive performance metrics; published a reproducible Quarto report online (manuscript in preparation).

SEP 2025 –
APR 2026

Mixed-Methods Analysis of Promotion and Prevention Mindsets

Undergraduate Research with Dr. Shane McCarty and 3 peers | Binghamton University

- Applied logistic regression modelling in R to investigate whether promotion and prevention mindsets predict health and safety-related goal preferences.
- Collaborated on a mixed-methods research project combining quantitative analyses with qualitative coding to examine public conceptions of community health and safety.
- Contributed to a 2025 SCRA presentation and served as first author on a poster abstract accepted for the 2026 APHA Annual Meeting.

AUG 2023 –
DEC 2024

First-Year Research Immersion: Community and Global Public Health

Undergraduate Research | Binghamton University

- Collaborated with a seven-person team on women's health, mental health, vaccination, and reproductive health research.
- Conducted SPSS analyses, designed and distributed a Qualtrics survey, and presented findings through two FRI poster presentations.

AUG 2023 -
DEC 2024

Political Affiliation and Zero-Sum Beliefs

Undergraduate Research with Dr. Shane McCarty and 2 peers | Binghamton University

- Examine political party affiliation and zero-sum beliefs as predictors of voter choice in the 2024 U.S. election.
- Conduct ANOVA/Kruskal-Wallis tests, logistic regression, and machine-learning classification in R (manuscript in preparation).

Professional Experience

JUN 2026 -
PRESENT

Quantitative Intern, First-Year Research Immersion Program

With Dr. Megan Fegley and Dr. Caitlin Light | Binghamton University

- Contribute to R Shiny dashboards for student outcomes and research dissemination metrics used in program evaluation and reporting.
- Develop an integrated dissemination database and Qualtrics surveys to collect publication and presentation outcomes across FRI programs.

JUN 2026 -
AUG 2026

Mental Health Workshop Quantitative Intern

Summer Internship with Dr. Shane McCarty | Binghamton University

- Designed Qualtrics surveys and quantitative evaluation methods assessing mental health literacy, well-being, nonviolent communication, and workshop effectiveness.
- Created interactive data visualizations and individualized wellness profiles using Observable JS.
- Developed an interactive wellness webpage integrating workshop materials, participant profiles, and trainer-facing visualizations for program evaluation.

Teaching & Mentoring Experience

JAN 2025 -
MAY 2025

R Programming Peer Mentor

First-Year Research Immersion Program | Binghamton University

- Mentored undergraduate students in foundational R programming, data analysis, code troubleshooting.
- Collaborated with the instructor and other research assistants to support research teams' Quarto research reports

Publications

McCarty, S., Light, C., Hei, Z., & Fegley, M. (2026).
FRI Peer Mentor Mindsets Guide. Binghamton University.
shanemccarty.github.io/FRI/mindsets

McCarty, S., & Hei, Z. (2025).
Selected chapters. In S. McCarty (Ed.), The Quantitative Playbook for Public Health Research in R.
shanemccarty.github.io/FRI/playbook/

Presentations & Conference Abstracts

2026	A Mixed-Methods Analysis of Promotion and Prevention Mindsets <i>American Public Health Association Annual (APHA) Meeting</i> Hei, Z., Giallella, J., & McCarty, S. First-author poster abstract accepted for presentation. https://apha.confex.com/apha/2026/meetingapp.cgi/Paper/594471
NOV 2025	Promotion & Prevention Mindsets: A Four-Study, Mixed-Methods Approach <i>Society for Community Research & Action (online)</i> Invited presentation; McCarty, S., Hei, Z., Rualo, G., Khalap, I., & Giallella, J.
DEC 2024	Public Knowledge & Attitudes on Vaccinations and Reproductive Health <i>FRI Proposal Poster Research Session 11th annual, Binghamton University</i>
DEC 2023	Endometriosis and Women's Mental Health <i>FRI Proposal Poster Research Session 10th annual, Binghamton University</i>

Data Science Projects

JAN 2026 - MAY 2026	Histopathology Thyroid Cancer Detection Using YOLO26 <i>Digital & Data Studies Group Project Binghamton University</i> Built a YOLO26-based deep-learning model to classify thyroid histopathology images and evaluated confusion matrices, Type I/II errors, class imbalance, and dataset representativeness.
SEP 2025 - DEC 2025	Deep Reinforcement Learning in Board Games <i>Digital & Data Studies Group Project Binghamton University</i> Implemented PyTorch reinforcement-learning agents for Tic-Tac-Toe and Connect Four through self-play and visualized training performance.

Awards, Honors & Scholarships

FEB 2026 - FEB 2027	NSF Research Experiences for Undergraduates (REU) Research Stipend Recipient <i>Binghamton University</i> Year-long research stipend supporting transcriptomic biomarker discovery and validation in chemo-immunotherapy with Dr. Nancy Guo.
FEB 2026	Pi Mu Epsilon <i>National Mathematics Honor Society</i>
FALL 2023 - PRESENT	Harpur College Dean's List <i>Binghamton University</i> Named to the Dean's List every semester attended.

Provost's Promise Scholarship

Binghamton University

- Recipient of renewable financial support each semester throughout undergraduate study.

Technical Skills

Programming: R (tidyverse, tidymodels, caret), Python (Pandas, NumPy, scikit-learn, PyTorch), SPSS

Data Visualization & Applications: ggplot2, Plotly, Matplotlib, Observable JS, R Shiny, Qualtrics

Research & Reproducibility: Quarto, GitHub, LaTeX

Methods: Regression, random forests, decision trees, MARS, deep learning, differential expression, GSEA, mixed-methods analysis

Communication & Collaboration: Public speaking; Team-based research in academic and internship settings
